Towards Data-Centric Counterfactual Analysis

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Abstract

Counterfactual analysis asks a fundamental question in data-driven decision making: what minimal and feasible changes to an input would result in a different outcome? Existing works are largely model-centric, generating synthetic modifications by probing predictive models, while many real-world applications require actionable and trustworthy answers from data. We articulate a vision of *data-centric counterfactual analysis*, where counterfactual reasoning is elevated to a data management primitive and formulated as a constrained, cost-aware and set-oriented retrieval problem. To realize this vision, we introduce *counterfactual querying*, which retrieves feasible, outcome-changing alternatives from observed data under explicit constraints, cost models, and set-level objectives. We develop a two-phase framework that integrates candidate generation that is based on constraints with set-level optimization, guided by the principles of interpretability, cost faithfulness, efficiency, and collective validity. We further present a prototype system, **DcCA**, demonstrating that counterfactual querying can be implemented efficiently across tabular, temporal, graph, text, and image data. Our results show that counterfactual querying goes beyond similarity search and enables structured, actionable reasoning over data. The source code and the full version of the paper, including the appendix, are available at https://anonymous.4open.science/r/DcCA-825B/.

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://anonymous.4open.science/r/DcCA-825B/.

1 Introduction

Counterfactual analysis [54, 80] asks a fundamental question in data-driven decision making: *"what minimal and feasible change would lead to a different outcome?"* By identifying semantically meaningful modifications that alter an outcome, it supports explanation [14], recourse [37], and auditing [45], with applications in healthcare [1], finance [84], and law [88]. Recent work further leverages counterfactuals to improve diversity [54], robustness [19], and fairness [79].

Despite this progress, existing approaches are largely *model-centric*, asking *how to modify an input to change a model's prediction*. They analyze gradients, perturb features, or generate synthetic edits by interrogating predictive models [6, 14, 25, 35, 38, 48, 54, 63, 73, 78]. Consequently, the resulting counterfactuals are often synthetic, tied to a specific model, may violate domain constraints, and need not correspond to observed data or feasible interventions [16, 49].

We argue that, particularly in high-stakes applications [1, 45, 84], counterfactuals should be grounded in observed data to ensure feasibility, verifiability, and interpretability. This shifts the question from generating hypothetical inputs to identifying *which real alternatives explain the outcome*. Rather than synthesizing new instances, we retrieve observed alternatives with different outcomes that are *close*, *feasible*, and *interpretable*, yielding a model-agnostic, evidence-based view of counterfactual analysis.

We envision **data-centric counterfactual analysis**, where counterfactual reasoning becomes a first-class data management primitive that retrieves feasible, outcome-changing alternatives from observed data under explicit constraints, costs, and objectives. This elevates counterfactual analysis from model interrogation to database query answering, placing it at the core of data management. The resulting query abstraction enables structured, constraint-aware reasoning over data and complements recent causal graph-based approaches [60], which support interventions and counterfactual queries through explicit causal models.

Our goal is not to develop another counterfactual generation algorithm, but to establish *counterfactual querying* as a first-class data management primitive. The challenge is therefore not generating a counterfactual, but designing the query abstraction, indexing mechanisms, optimization strategies, and execution model for counterfactual reasoning as a recurring database operation.

From this perspective, counterfactual querying differs fundamentally from existing retrieval pipelines. While existing systems compose vector search, filtering, reranking, and subset selection to retrieve similar objects, counterfactual querying asks *which feasible interventions would change the outcome?* Accordingly, the opposite-outcome predicate and admissibility define the feasible search space, intervention cost ω replaces geometric similarity as the optimization objective, and the answer is a set of complementary counterfactuals that jointly optimizes a user-specified objective Ω.

These semantics cannot be captured by composing existing retrieval operators. Treating retrieval, feasibility, intervention cost,

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and set selection as separate stages prevents holistic optimization and obscures the query semantics. We therefore elevate the opposite-outcome predicate, intervention cost ω, and set objective Ω to first-class query constructs, enabling future data systems to optimize counterfactual reasoning as a native database operation.

Realizing this vision requires addressing two key challenges: (i) supporting counterfactual queries efficiently and at scale, and (ii) enabling principled reasoning across heterogeneous data modalities.

1.1 Key Challenge 1: Efficient and Scalable Counterfactual Query Answering

Counterfactual queries resemble similarity search in that both retrieve instances close to a query under a distance metric [10, 32, 51]. A natural approach is therefore to retrieve nearest neighbors with different outcomes. However, users seek not merely similar instances, but *feasible, outcome-changing* alternatives that provide *actionable* insights [8, 47, 52, 75]. Counterfactual querying must therefore enforce *label constraints*, respect *admissibility conditions*, optimize *cost-faithful intervention metrics*, and return *informative rather than redundant* answers [12].

These requirements fundamentally distinguish counterfactual querying from nearest-neighbor search. Feasibility is governed by often non-metric domain constraints, intervention cost reflects real-world effort rather than geometric proximity, and selecting a small set of counterfactuals requires jointly optimizing objectives such as diversity and fairness. Addressing these challenges calls for new representations, indexing methods, constraint-aware retrieval, and set-level optimization.

1.2 Key Challenge 2: Counterfactual Querying Across Modalities

Real-world data is inherently multimodal, combining tables, graphs, sequences, text, and images, each with distinct structures, such as attributes, temporal dependencies, and graph topology. Although existing systems learn effective shared representations across modalities, this embedding-centric paradigm remains limited for structured querying and reasoning, particularly when enforcing constraints or composing information. The limitation is especially pronounced for counterfactual reasoning, where retrieval must satisfy structured feasibility conditions rather than similarity alone [80].

We advocate a data-centric view that models each modality through its entities, relationships, dependencies, and constraints. Counterfactual queries therefore operate on *semantic dimensions of change* rather than raw features. These dimensions differ across modalities—attributes in tables, temporal signals in sequences, and structures in graphs—and may have different manifestations across modalities. Consequently, similarity search alone may return candidates that violate constraints or lack semantic validity. Moreover, robustness [29], diversity [44], and fairness [43] introduce dependencies among candidates, requiring principled methods that jointly optimize feasibility, interpretability, and set-level utility across modalities.

In summary, we present a data-centric vision of counterfactual analysis and formalize it as *counterfactual querying*, a first-class data management primitive with a query model, a two-phase framework, and core design principles.

The remainder of the paper is organized as follows. Section 2 introduces the query model and framework. Section 3 studies instance-level semantics across modalities. Section 4 presents set-level optimization for collective validity. Section 5 examines existing system limitations and presents the proof-of-concept system DcCA. Section 6 concludes.

2 Answering Counterfactual Queries

This section formalizes counterfactual querying and presents a framework for answering such queries, providing the foundation for the proposed data-centric paradigm.

2.1 Counterfactual Queries

We model counterfactual querying as a first-class data management abstraction. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ be a labeled dataset, where $\mathbf{x}_i \in \mathbb{R}^d$ is an instance with d attributes and y_i its outcome (or class label). Given a query $q = (\mathbf{x}_q, y_q)$, a *counterfactual query* retrieves a size-k set of instances from $\mathcal{D}$ that (i) have outcomes different from y_q, (ii) are close to $\mathbf{x}_q$ under a suitable notion of change, and (iii) satisfy domain constraints. These instances provide realistic and feasible alternatives. For example, income may be changeable whereas age is immutable, and edits must preserve validity. Among feasible candidates, we favor those requiring minimal change while collectively providing diverse or informative alternatives. Formally, a counterfactual query is specified by (q, ω, Ω, k), where ω defines instance-level feasibility and transformation cost, and Ω evaluates the quality of the returned set.

The specification ω jointly defines admissibility and the cost of transforming $\mathbf{x}_q$ into a candidate $\mathbf{x} \in \mathbb{R}^d$. It induces a cost $\omega(\mathbf{x}_q, \mathbf{x}) \in \mathbb{R}_{\geq 0} \cup \{\infty\}$, where $\omega(\mathbf{x}_q, \mathbf{x}) < \infty$ if and only if the transformation is admissible. The feasible search space is

$$\omega(q) = \{\mathbf{x}_i : (\mathbf{x}_i, y_i) \in \mathcal{D},\ y_i \neq y_q,\ \omega(\mathbf{x}_q, \mathbf{x}_i) < \infty\}. \tag{1}$$

In practice, ω is instantiated per modality and deployment. For tabular data, it is a weighted sum of per-column monotonic edit costs with ∞ assigned to immutable attributes (e.g., age); for temporal data, it combines an alignment cost with a dynamics-validity predicate; for text and image, it combines a representation-space cost with a semantic-admissibility predicate (Section 3). Operationally, ω is *defined* from domain or policy rules, optionally *learned* from logged interventions or expert preferences, and *validated* against held-out feasible outcomes or expert audit. Composite constraints are conjunctive, so violating any constraint yields $\omega = \infty$, cleanly separating feasibility from intervention effort.

The objective Ω evaluates the collective quality of a candidate set $S \subseteq \omega(q)$, capturing preferences such as diversity, fairness, and robustness. Counterfactual querying therefore balances set quality under Ω with transformation cost under ω, yielding

$$S^\star \in \arg\max_{S \subseteq \omega(q), |S|=k} \left\{ \Omega(S; q) - \sum_{\mathbf{x} \in S} \omega(\mathbf{x}_q, \mathbf{x}) \right\}. \tag{2}$$

Evidence availability and query semantics. Data-centric counterfactual querying operates over observed data, so its answers are bounded by the evidence in $\mathcal{D}$. If $\omega(q) = \emptyset$, no feasible outcome-changing instance satisfies the specified constraints, and the correct answer is the empty set. Rather than indicating failure, the empty

set certifies that no evidence-supported counterfactual exists under ω. Likewise, if $0 < |\omega(q)| < k$, the query returns all feasible counterfactuals together with their provenance. In both cases, the result faithfully reflects the available evidence.

This evidence-based semantics distinguishes data-centric from model-centric and generative counterfactual analysis. Data-centric querying asks *"What feasible alternatives are supported by the observed data?"* and emphasizes realism, feasibility, provenance, and verifiability. Model-centric methods instead ask *"What alternative would a predictive model produce?"* and can always synthesize a counterfactual without guaranteeing that it is observed or feasible. The two paradigms therefore answer different questions rather than serving as fallbacks for one another. As $\mathcal{D}$ grows, the coverage of data-centric counterfactual querying naturally increases while preserving its evidence-based semantics.

2.2 A Two-Phase Query Answering Framework

To realize counterfactual querying in a data system, we develop a two-phase framework that combines instance-level semantics ω (feasibility and cost) with set-level optimization under Ω. The framework consists of an offline phase that prepares reusable structures over $\mathcal{D}$ and a specification space $\mathcal{W}$, followed by an online phase that efficiently answers queries (q, ω, Ω, k).

Offline (Preprocessing for $(\mathcal{D}, \mathcal{W})$) Given $\mathcal{D}$ and a *specification family* $\mathcal{W}$, i.e., a set of specifications ω sharing the same structural primitives—immutable attributes, constraint types, and cost components—and therefore admitting a common index while differing only in concrete parameters (e.g., per-attribute weights or thresholds), the system builds reusable structures for evaluating feasibility and cost under different $\omega \in \mathcal{W}$. First (**Step 1.a**), it identifies the structural primitives defining admissibility and transformation cost. Second (**Step 1.b**), it constructs a constraint-aware index $\mathcal{I}$ over $\mathcal{D}$ organized by these primitives.

The design of $\mathcal{I}$ depends on ω. When ω primarily reweights dimensions, metric-aware structures such as weighted KD-trees [70], ball trees [18], and cover trees [33] support efficient pruning. When ω is dominated by discrete constraints, partitioned indices [46], bitmap filtering [9], and inverted indices [4] efficiently prune infeasible candidates before cost evaluation.

Online (querying over (q, ω, Ω, k)). Directly optimizing Eq. (2) is expensive. We therefore adopt a coarse-to-fine strategy over $\mathcal{I}$.

First (**Step 2.a**), the system retrieves a candidate pool $\mathcal{A}_\omega(q)$ containing admissible instances with low transformation cost, returned in nondecreasing cost order for early pruning. If $\omega \notin \mathcal{W}$, it falls back to a linear scan over $\mathcal{D}$. Formally,

$$\mathcal{A}_\omega(q) \in \arg \min_{\substack{\mathcal{A}_\omega \subseteq \omega(q) \\ |\mathcal{A}_\omega| = m}} \left\{ \sum_{\mathbf{x} \in \mathcal{A}_\omega} \omega(\mathbf{x}_q, \mathbf{x}) \right\}, \tag{3}$$

where $m \geq k$ controls the candidate-pool size.

Next (**Step 2.b**), it selects the final answer $S^\star \subseteq \mathcal{A}_\omega(q)$ by optimizing Eq. (2). Finally (**Step 2.c**), it returns each $\mathbf{x} \in S^\star$ together with an interpretable edit $\Delta(\mathbf{x}_q, \mathbf{x})$ consistent with ω.

The coarse-to-fine design bounds the cost of constraint enforcement and feasibility validation. Cheap attribute- and metric-level checks are performed during candidate generation (Step 2.a), pruning most infeasible instances before expensive validation. Execution-level (temporal, graph) and semantic (text, image) validation is then applied only to the candidate pool $\mathcal{A}_\omega(q)$, so validation scales with $|\mathcal{A}_\omega|$ rather than $|\mathcal{D}|$. Because candidates are processed in nondecreasing cost order, validation terminates once k admissible answers are found. Increasing m improves the likelihood of finding k feasible counterfactuals at the cost of additional validation, providing a controllable accuracy–latency trade-off (Principle P3 in Section 2.3).

2.3 Core Principles

The query formulation above gives rise to four principles for designing counterfactual query systems: interpretability, cost faithfulness, efficiency, and collective validity.

P1 (Interpretability). *Counterfactual results must be interpretable in the original domain.* For each selected counterfactual $\mathbf{x} \in S^\star$, the system returns a domain-level edit $\Delta(\mathbf{x}_q, \mathbf{x})$ (e.g., feature changes in tabular data or valid edit sequences in structured data). The representation and index must preserve sufficient information to reconstruct $\Delta(\mathbf{x}_q, \mathbf{x})$, and the cost $\omega(\mathbf{x}_q, \mathbf{x})$ must be grounded in this edit, ensuring actionable rather than similarity-based results.

P2 (Cost faithfulness). *The cost function $\omega(\mathbf{x}_q, \mathbf{x})$ must reflect intervention effort.* If $\omega(\mathbf{x}_q, \mathbf{x}_a) < \omega(\mathbf{x}_q, \mathbf{x}_b)$, then transforming $\mathbf{x}_q$ into $\mathbf{x}_a$ should require no more effort than transforming it into $\mathbf{x}_b$ under the intended domain model. Infeasible transformations are assigned infinite cost, cleanly separating feasibility from effort.

P3 (Efficiency). *The system must efficiently retrieve admissible candidates in $\omega(q)$ at scale.* It should provide predictable performance, bounded resource usage, and controllable accuracy–latency trade-offs, enabling practical deployment over large and heterogeneous datasets.

P4 (Collective validity). *The returned answer set must jointly optimize transformation cost and set-level objectives.* Given (q, ω, Ω, k), the solution $S^\star$ satisfies $|S^\star| = k$ and optimizes Eq. (2). Since Ω captures collective objectives such as diversity, robustness, and fairness, the system must balance them against per-instance cost to construct informative and complementary answer sets.

Together, P1–P4 characterize counterfactual querying as structured, constraint-aware, and set-oriented retrieval rather than similarity search.

2.4 An Illustrative Example on Tabular Data

We illustrate counterfactual querying using a loan approval scenario. Suppose Maya applies for a loan and is denied. Her record is `(Income=$58k, Util.=41%, Delinq.=1, State=NC, Age=28)`. She asks: *"What would I need to change to get approved?"* This naturally defines a counterfactual query (q, ω, Ω, k), where ω specifies feasibility and intervention cost (e.g., age is immutable and attributes incur different costs), k is the number of returned alternatives, and Ω encourages diverse recourse strategies.

Offline. The system constructs a label-aware index $\mathcal{I}$ over $\mathcal{D}$ using ω to encode admissibility and cost (e.g., Age $\neq 28$ is infeasible), supporting interpretable edits, cost-faithful ranking, and efficient retrieval (P1–P3).

Online. Given (q, ω, Ω, k), the system retrieves a low-cost candidate pool, selects the final answers by optimizing Eq. (2), and

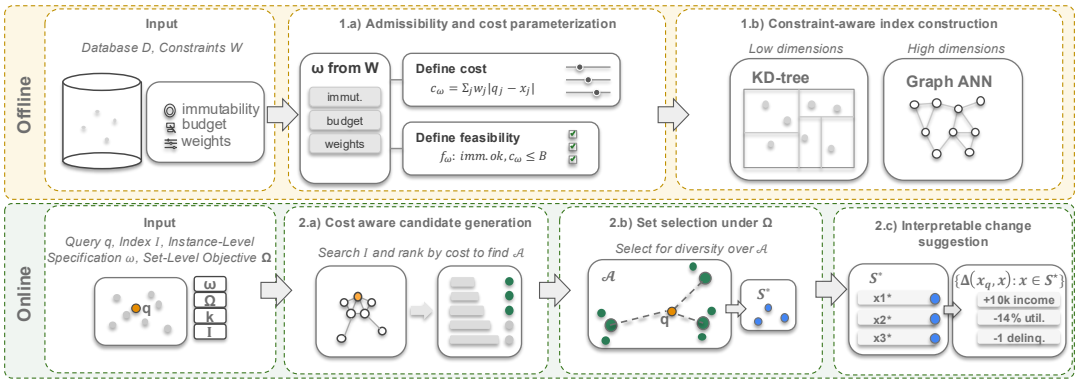


Figure 1: Two-phase counterfactual query pipeline.

returns each counterfactual together with its interpretable edit $\Delta(\mathbf{x}_q, \mathbf{x})$.

Maya may receive three complementary paths to approval, such as (i) `+10k income with -2% utilization`, (ii) `-14% utilization`, or (iii) `-1 delinquency with -6% utilization`. These results are grounded in observed data (P1), feasible under ω (P2), efficiently retrieved (P3), and collectively optimized under Ω (P4), illustrating data-centric counterfactual reasoning.

3 Instance-Level Semantics Across Modalities

Building on the query model, this section examines how to realize instance-level semantics across data modalities. The central challenge is to retrieve candidates that satisfy feasibility constraints while remaining competitive under the transformation cost induced by ω. Depending on the modality, admissibility may be enforced through explicit attributes, structural constraints, or execution- and semantics-level validation. We examine how P1 (interpretability), P2 (cost faithfulness), and P3 (efficiency) guide counterfactual querying across tabular, temporal, graph, text, and image data.

Tabular and Relational Data. Tabular data provides the clearest instance-level semantics. Since features are represented column-wise, edits are directly interpretable: $\Delta(\mathbf{x}_q, \mathbf{x})$ is obtained by comparing columns, $\omega(\mathbf{x}_q, \mathbf{x})$ aggregates per-column intervention costs, and immutable attributes are enforced by assigning infinite cost. Consequently, tabular data naturally satisfies P1 and P2.

Nevertheless, counterfactual querying remains fundamentally different from nearest-neighbor search. Columns differ in semantic scale and actionability, making geometric distance a poor proxy for intervention cost [36]. Moreover, immutable attributes and structural rules partition the search space into constrained subspaces [24], so distance alone cannot guarantee feasibility [5]. Effective systems must therefore incorporate column-level semantics into indexing and candidate generation.

Sequence and Temporal Data. For temporal data, $\mathbf{x}_q$ is a trajectory $(\mathbf{x}_{q,1}, \ldots, \mathbf{x}_{q,T})$. Transformation cost is commonly estimated using alignment measures such as dynamic time warping (DTW) [74] or learned sequence embeddings [55]. However, admissibility is governed by the underlying dynamics rather than alignment alone. For example, two similar chess move sequences may lead to different validities because each move depends on the evolving board state.

This separation between similarity and feasibility raises three challenges. Edits must correspond to valid trajectory transformations [34]; alignment may poorly approximate intervention cost because of time shifts or scaling [68]; and early edits propagate through the sequence, making feasibility path-dependent and requiring execution-level validation [61]. Counterfactual querying therefore requires dynamics-aware constraints beyond sequence similarity.

Text, Image, and Graph Data. Text and image retrieval similarly separates scalable retrieval from semantic admissibility. In text, surface metrics such as edit distance or BLEU [62] poorly capture semantic change, so ω is typically approximated in pretrained representation spaces [17]. Yet admissibility depends on higher-level semantics, where local edits may alter global meaning [86], requiring full-sequence reasoning or external grounding [27, 82].

Image retrieval likewise relies on embeddings, but pixel-level changes rarely correspond to meaningful semantic edits [56]. This weakens traceability [3], misaligns perceptual similarity with intervention cost [90], and often requires localization or scene-level validation [23]. Concept bottleneck models partially bridge this gap by introducing interpretable semantic representations [15, 20, 42].

In graph data, counterfactuals correspond to structural edits over nodes and edges [26]. Retrieval therefore relies on embeddings for scalability [41], whereas admissibility depends on structural constraints such as connectivity, typing, or capacity. As in temporal data, similarity is geometric while validity is structural [39], creating embedding–constraint misalignment [2] and requiring costly verification [83]. Effective graph counterfactual querying must therefore integrate structural constraints into retrieval.

Extended to Multi-Modality. When instances span multiple modalities, counterfactual querying requires coordinated reasoning across views. A candidate must satisfy modality-specific constraints while remaining cross-modally consistent [11]. Consequently, traceability (P1), cost faithfulness (P2), and efficiency (P3) must all extend across heterogeneous representations, while shared semantic factors may appear differently in different modalities.

Similarity search over joint or modality-specific embeddings therefore serves only as a proposal mechanism; correctness is established through cross-modal validation that enforces feasibility and consistency. Counterfactual querying is thus inherently verification-driven, providing a unified foundation for scalable reasoning across heterogeneous data.

Table 3 in Appendix A summarizes the constraint type, edit representation, and enforcement point for each modality. For multimodal queries, ω additionally includes a cross-modal consistency predicate: a candidate must satisfy each modality's constraints *and* remain jointly consistent across modalities [11] (e.g., a tabular attribute edit must imply consistent changes in the associated text and image). The current prototype enforces these constraints through validation after shared-embedding retrieval. More importantly, data-centric counterfactual analysis exposes them as *declarative query semantics*, enabling future optimizers to push modality-specific and cross-modal constraints into indexing and retrieval, analogous to the evolution of relational systems from post-filtering to predicate-aware query optimization. Achieving this, rather than relying on post-filtering, is a central open challenge of this vision.

4 Collective Validity in Counterfactual Queries

While instance-level semantics define feasibility and transformation cost through ω, counterfactual querying is inherently set-valued: users seek a small set of complementary alternatives. We therefore introduce set-level optimization to construct informative answer sets, extending retrieval to structured subset selection over diversity, robustness, and fairness.

This is captured by the principle of collective validity (P4), where answer quality depends not only on individual feasibility and cost but also on interactions among returned instances. Simply returning the top-k lowest-cost candidates often yields redundant answers. Instead, a set-level objective Ω balances transformation cost with collective utility.

Robustness. Robust counterfactuals should remain stable under small perturbations of the query [29]. Let $\mathcal{N}(\mathbf{x}_q)$ denote a neighborhood of the query instance. A robustness objective is

$$\Omega_{\text{rob}}(S) = -\mathbb{E}_{\mathbf{x}'_q \sim \mathcal{N}(\mathbf{x}_q)} \left[\min_{\mathbf{x} \in S} d(\mathbf{x}'_q, \mathbf{x}) \right]. \tag{4}$$

Maximizing $\Omega_{\text{rob}}(S)$ encourages the selected set to remain representative under local perturbations, producing stable and trustworthy recommendations. It therefore promotes collective validity by preserving utility across plausible query variations.

Diversity. Counterfactuals should expose distinct paths of change rather than minor variations along a single dimension [44]. A diversity objective is

$$\Omega_{\text{div}}(S) = \sum_{\mathbf{x}_i, \mathbf{x}_j \in S,\, i<j} d(\mathbf{x}_i, \mathbf{x}_j). \tag{5}$$

Maximizing $\Omega_{\text{div}}(S)$ encourages complementary rather than redundant alternatives, revealing multiple recourse strategies. This formulation corresponds to the max-sum diversification problem [7], for which efficient approximation algorithms exist.

Fairness. For auditing and decision support, the composition of the returned set is critical [43]. Let $g(\mathbf{x})$ denote the protected attribute of instance $\mathbf{x}$ (e.g., gender or ethnicity), and let π_a denote the desired proportion of group a in the returned set. A fairness objective is

$$\Omega_{\text{fair}}(S) = -\left| \frac{1}{|S|} \sum_{\mathbf{x} \in S} \mathbf{1}[g(\mathbf{x}) = a] - \pi_a \right|. \tag{6}$$

Maximizing $\Omega_{\text{fair}}(S)$ encourages the selected set to match the desired group proportions, reducing bias while introducing global dependencies among candidates. In practice, fairness must be balanced against objectives such as cost and diversity.

5 DcCA: A Proof-of-Concept System

To demonstrate the feasibility of our vision, we present **DcCA** (Data-centric Counterfactual Analysis), a proof-of-concept system that realizes counterfactual querying as a first-class data management primitive. Building on the query model, principles (P1–P4), and the two-phase framework (Sections 2–4), **DcCA** integrates constraint-aware retrieval, cost-faithful reasoning, and set-level optimization into a unified system for structured, actionable counterfactual analysis. We first identify the limitations of existing systems and then show how **DcCA** addresses them through a principled design and preliminary empirical results.

5.1 Limitations in Existing Work

Modern database systems increasingly integrate vector and similarity search, including PostgreSQL (pgvector) [64], Neo4j [57], Snowflake [76], and Oracle [58, 59]. They support scalable nearest-neighbor search together with structured filtering, hybrid lexical–semantic matching [28, 85], and reranking [40]. However, these capabilities remain centered on similarity search and do not jointly support the constraint-aware, cost-faithful, and set-oriented semantics required by counterfactual querying.

A fundamental limitation is their pipeline-based design, which separates retrieval, filtering, and reranking [13, 53]. This conflicts with counterfactual querying because feasibility constraints and opposite-outcome predicates reshape the search space rather than acting as post-filters, while intervention costs (P2) generally differ from the geometric metrics assumed by existing indexes. Consequently, systems may return infeasible candidates or rely on post hoc validation, compromising both efficiency (P3) and correctness.

Current retrieval systems expose vector search, filtering, and subset selection as independent operators. As a result, counterfactual queries can only be executed as staged pipelines, where feasibility is checked after retrieval and set-level objectives are optimized after candidate generation. This decomposition prevents joint optimization of access paths, indexes, and execution plans with respect to the opposite-outcome predicate, intervention cost ω, and set objective Ω. Table 1 summarizes this gap: existing systems support these capabilities individually, whereas counterfactual querying requires them as first-class query constructs.

These limitations are evident in practice. For example, **pgvector** applies filtering after index traversal [64], reducing valid candidates and often requiring repeated scans. Approximate nearest-neighbor methods exhibit similar behavior, as post-filtering under structured constraints can degrade recall and stability [21, 50]. More fundamentally, current systems do not treat feasibility specifications ω, counterfactual predicates, or set-level objectives Ω as first-class

Table 1: Capabilities required by counterfactual querying vs. existing paradigms; only DcCA provides all jointly. *Opposite*: opposite-outcome predicate; *Cost*: non-metric, asymmetric transformation cost ω; *Feasib.*: feasibility reshapes the search space (not a post-filter); *Set*: set-level objective Ω; *Edit*: interpretable edit Δ.

Paradigm	Opposite	Cost	Feasib.	Set	Edit
Constrained retrieval [46]	No	No	Partial	No	No
Vector DB / ANN [50, 64]	No	No	No	No	No
NN search [32]	No	No	No	No	No
Subset selection [7]	No	No	No	Yes	No
FACE [65]	Yes	Partial	Partial	No	Partial
DcCA (ours)	Yes	Yes	Yes	Yes	Yes

Table 2: Runtimes across modalities comparing DcCA with state-of-the-art baselines. N/A indicates unsupported stages. All times are in milliseconds (ms): preprocessing and indexing are measured over the dataset, while search (Step 2.a) and set selection (Step 2.b) are reported per query.

Method	Dim	Feature	Engine	Retrieval	Prep.	Index	Search	Set
Tabular (N=10000)								
Baseline	20	Raw	DuckDB	Linear Scan	0.00	23.62	2.66	N/A
KD-Tree	20	Raw	SciPy	KD-Tree Search	0.00	3.57	0.98	N/A
DcCA	64	Table2Vec	FAISS	k-NN Search	14.03	0.87	3.91	3.03
Temporal (N=500)								
Baseline	50	Raw	tslearn	DTW Scan	0.00	0.00	2377.50	N/A
SAX	10	Symbolic	iSAX	Hamming Search	0.93	0.00	0.49	N/A
DcCA	128	Time2Vec	FAISS	k-NN Search	2.05	0.06	1.28	2.51
Text (N=500)								
Baseline	512	Keyword	DuckDB-FTS	Keyword Match	0.00	2.75	1.12	N/A
BM25	512	BM25	Whoosh	BM25 Ranking	0.00	29.25	14.67	N/A
DcCA	384	SBERT	FAISS	k-NN Search	253.01	0.09	1.57	2.82
Image (N=100)								
Baseline	150528	Raw	DuckDB-Blob	Linear Scan	0.00	0.00	103.75	N/A
dHash	64	dHash	FastBit	Hamming Search	0.09	0.00	0.09	N/A
DcCA	512	ResNet18	FAISS	k-NN Search	1921.81	0.74	3.69	3.20
Graph (N=2000)								
Baseline	N/A	Adjacency	Neo4j	BFS Search	0.00	6.30	0.17	N/A
DcCA	128	GCN	Neo4j-VEC	k-NN Search	3.73	0.14	1.52	2.60
MultiModal: Tabular + Text + Image (N=100)								
DcCA	512	CLIP	FAISS	k-NN Search	2265.25	0.14	2.32	2.82

query constructs. Consequently, they cannot jointly support traceable edits (P1), cost-faithful ranking (P2), constraint-aware retrieval (P3), and collective validity (P4).

Bridging this gap requires elevating counterfactual querying to a first-class data management abstraction, enabling retrieval, constraints, intervention costs, and set-level objectives to be specified and optimized jointly.

5.2 DcCA and Preliminary Results

To demonstrate the feasibility of our vision, we implement **DcCA**, a prototype system that instantiates our query model, principles, and two-phase framework. **DcCA** supports tabular, temporal, graph, text, image, and multimodal data. This makes **DcCA** the only evaluated system that supports unified counterfactual querying over heterogeneous modalities, enabling constraint-aware retrieval (P1–P3) and set-level optimization (P4) in a single pipeline.

DcCA employs a shared embedding-based design for efficient cross-modal processing, using `Table2Vec` [89], `Time2Vec` [39], `GCN` [41], `SBERT` [71], and `ResNet18` [30]. For multimodal inputs, **DcCA** represents tabular, text, and image information in a unified embedding space via `CLIP` [67]. Following our framework, Step 2.a performs cost-aware retrieval using `FAISS` [22] and `Neo4j-VEC` for graphs, while Step 2.b performs set-level selection under Ω. All experiments are conducted on a Mac M2 Pro with 32 GB RAM.

To validate the efficiency of our design, we compare **DcCA** with representative system pipelines, including `DuckDB` [66] for relational filtering, `KD-Tree` [69] for tabular nearest-neighbor search, `tslearn` (DTW) [77] and `SAX` [87] for temporal matching, `Neo4j` [57] for graph traversal, `DuckDB-FTS` [66] and `BM25` [72] for text retrieval, and both linear scan [31] and hashing-based methods [81] for image retrieval. As shown in Table 2, **DcCA** achieves substantial efficiency gains while supporting richer query semantics. For example, on temporal data ($N = 500$), DTW-based retrieval requires 2377.50 ms, whereas **DcCA** completes candidate retrieval in 1.28 ms, with set-level optimization adding only 2.51 ms. In the multimodal setting, **DcCA** supports joint tabular, text, and image counterfactual querying, completing candidate retrieval in 2.32 ms and set-level selection in 2.82 ms.

These results highlight two key advantages. First, **DcCA** integrates constraint-aware retrieval and cost-faithful ranking, avoiding the inefficiencies of post hoc filtering. Second, unlike modality-specific pipelines, **DcCA** supports set-level reasoning across both single-modality and multimodal inputs, producing diverse and interpretable counterfactual sets in real time. Overall, the prototype shows that counterfactual querying can be realized efficiently across heterogeneous data modalities, bridging current system capabilities with the requirements of data-centric counterfactual analysis.

Beyond runtime, we further study counterfactual *quality*—collective-validity objectives, a comparison against representative counterfactual generators (FACE, DiCE, Wachter, GeCo), and approximation/scalability—together with the practical index-construction and maintenance costs; for space, these are reported in Appendix A. In brief, objective-specific set selection improves every collective-validity objective over top-k (which is best for none), and **DcCA** uniquely returns feasible, in-distribution counterfactuals whereas model-centric generators are infeasible and off-manifold.

6 Conclusions

We introduced *counterfactual querying* as a new data management primitive: a constrained, cost-aware, and set-oriented retrieval problem over observed data. Guided by interpretability (P1), cost faithfulness (P2), efficiency (P3), and collective validity (P4), we developed a query model, a two-phase framework, and a systems perspective that elevates counterfactual reasoning from model interrogation to database query processing. Our analysis across data modalities and the **DcCA** prototype demonstrate both the feasibility of data-centric counterfactual querying and the limitations of existing similarity-based retrieval systems. Success criteria are summarized in Appendix B.

Looking ahead, elevating counterfactual querying to a first-class abstraction opens opportunities to integrate constraints, intervention costs, and decision objectives directly into query processing. Advancing this vision requires new techniques for constraint-aware indexing, cost modeling, query optimization, and set-level reasoning, enabling future data systems to support structured, interpretable, and actionable reasoning for human-centered AI.

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A Additional Evaluation

Here, we provide a detailed counterfactual-*quality* study of DcCA.

Modality-specific semantics. Table 3 lists, for each modality, the constraint type that defines admissibility, the interpretable edit Δ, and the stage at which the constraint is enforced (referenced from Section 3). Feasibility shifts from cheap, index-time checks for tabular data to progressively more expensive execution-level, structural, and semantic validation for temporal, graph, text, and image data—precisely the gradient the coarse-to-fine pipeline of Section 2 exploits by applying costly checks only to the small candidate pool.

Table 3: Modality-specific counterfactual semantics: constraint type, edit representation, and where each constraint is enforced.

Modality	Constraint type	Edit Δ	Enforced at
Tabular	column rules, immutability	attribute deltas	index / candidate gen.
Temporal	dynamics, execution validity	valid sub-trajectory	execution validation
Graph	connectivity, typing, capacity	structural edits	structural validation
Text	semantic admissibility	span rewrite	semantic validation
Image	scene/region semantics	localized edit	semantic validation

Collective validity. Table 4 evaluates the set objectives of Section 4—diversity Ω_{div}, robustness Ω_{rob}, and fairness Ω_{fair}—for top-k lowest-cost selection versus objective-specific Set-step selectors (k=5, pool m=100, 200 queries). Each objective is best served by its own selector and top-k is optimal for none: on tabular data diversity rises from 97.12 to 117.60, robustness from −14.16 to −14.09, and fairness from −0.179 to −0.100, while on temporal data diversity optimization increases Ω_{div} more than six-fold (151.52 → 957.31). The gains trade off against cost (e.g., temporal cost 14.86 → 77.09) and against one another, so collective validity (P4) cannot be reduced to lowest-cost top-k and requires a first-class, user-specified Ω.

Table 4: Collective-validity objectives (Section 4) for top-k vs. objective-specific Set-step selection; higher Ω is better, lower Cost is better. Best per column in bold.

Setting	Selector	$\Omega_{\text{div}}\uparrow$	$\Omega_{\text{rob}}\uparrow$	$\Omega_{\text{fair}}\uparrow$	Cost$\downarrow$
Tabular	Top-k	97.12	−14.16	−0.179	**9.07**
	Div-opt	**117.60**	−14.44	−0.196	9.96
	Rob-opt	100.46	−14.09	−0.197	9.20
	Fair-opt	110.18	−14.33	−0.100	9.68
Temporal	Top-k	151.52	−21.62	−0.190	**14.86**
	Div-opt	**957.31**	−22.15	−0.185	77.09
	Rob-opt	153.08	−21.59	−0.186	14.91
	Fair-opt	824.42	−21.90	−0.100	61.80

Counterfactual capability. Table 5 compares DcCA with reference reimplementations of representative counterfactual generators on validity, feasibility (immutability), in-distribution rate, diversity, and cost. Nearly all methods flip the label (validity ≈ 100%), so the decisive columns are feasibility and in-distribution rate: model-centric generators (DiCE, Wachter, GeCo) are 0% feasible and essentially off-manifold, and FACE respects immutability only 1.9% of the time, whereas DcCA is uniquely both 100% in-distribution and substantially feasible (64.7%) and returns the most diverse valid set (Ω_{div}=27.08). The generators' lower nominal cost (1.73–1.89 vs. 4.09) is obtained precisely through infeasible, off-manifold edits—the complementary trade-off that DcCA resolves in favor of feasibility, realism, and verifiability.

Table 5: Counterfactual capability: DcCA vs. model-centric generators. Controlled simulation on synthetic tabular data (N=6000, 3 categorical immutable attributes, k=5, 150 queries) with reference reimplementations of each method's core mechanism—not the released libraries and not UCI data. Higher is better except Cost.

Method	Valid.%	Feas.%	In-dist.%	Ω_{div}	Cost
DcCA (data-centric)	100.0	**64.7**	**100.0**	**27.08**	4.09
FACE (data-path) [65]	100.0	1.9	100.0	17.29	2.83
DiCE (diverse grad) [54]	100.0	0.0	1.1	9.32	1.73
Wachter (grad) [80]	100.0	0.0	0.0	2.27	**1.89**
GeCo (genetic) [73]	80.8	0.0	0.7	20.24	1.77

Approximation and scalability. Tables 6 and 7 characterize efficiency. Approximate IVF-style retrieval trades speed for recall gracefully: nprobe=8 already attains perfect recall@100 at a 5.6× speedup over exact search, confirming that standard approximate-nearest-neighbor techniques transfer to counterfactual retrieval over embeddings. As the corpus grows to N=5 × 10^5, set selection stays essentially constant (≈ 1.5 ms) because it operates only on the small candidate pool, while brute-force search grows roughly linearly (0.10 → 129.09 ms) and dominates latency—identifying retrieval, not set optimization, as the scaling bottleneck and motivating the constraint-aware index of Section 2.

Table 6: Exact vs. approximate (IVF-style) retrieval; N=10000, d=64, k=100. Exact search: 1.65 ms/query.

nprobe	Recall@100	Speedup vs. exact
1	0.425	22.7×
4	0.940	9.4×
8	1.000	5.6×
16	1.000	2.7×

Table 7: Scalability of online query answering (search + set selection), d=64. Set selection is near-constant; retrieval dominates scaling.

N	Search (ms)	Set-step (ms)	Total (ms)
1,000	0.10	1.47	1.57
10,000	1.25	1.49	2.74
100,000	15.40	1.46	16.87
500,000	129.09	1.51	130.60

Practical realization and costs. The offline index is amortized across queries: the specification family $\mathcal{W}$ is partitioned into a small number of structural families (metric-reweighting vs. discrete-constraint dominated), each served by one reusable index, so construction cost scales with the number of families rather than $|\mathcal{W}|$. Under insertions and deletions to $\mathcal{D}$ the index is maintained incrementally, and a query whose $\omega \notin \mathcal{W}$ falls back to a linear scan, so the cost model degrades gracefully.

B Success Criteria

To realize our vision of data-centric counterfactual analysis, success should be measured not only by performance, but by how well systems support the semantics, principles (P1–P4), and design goals established in Sections 1–5. We outline five criteria that characterize a successful counterfactual query system.

C1 (Semantic queryability). Counterfactual querying should enable *action-oriented* questions (e.g., "what needs to change?") rather than predictive queries alone. This extends data systems from passive retrieval to structured reasoning over constraints, costs, and outcomes.

C2 (Unified data–model integration). Systems should treat data, constraints, and learned representations as interoperable components. Rather than relying solely on embeddings or model introspection, they should integrate structured semantics with learned features within a unified query framework.

C3 (Constraint-, cost-, and efficiency-aware processing). Systems must natively support feasibility constraints, cost-faithful transformations (P1–P2), and efficient query execution at scale (P3). This requires integrating constraint-aware indexing, cost-aware retrieval, and scalable query processing, rather than relying on post hoc filtering.

C4 (Set-level reasoning). Systems should elevate set-level objectives (e.g., diversity, robustness, fairness) to first-class query constructs (P4). This enables principled selection of complementary counterfactuals, moving beyond tuple-at-a-time retrieval to holistic answer construction.

C5 (Actionable outputs). Counterfactual queries should return interpretable and executable edits grounded in real data. This ensures that results are not only correct but also actionable and trustworthy, supporting human-in-the-loop decision making.

Together, these criteria define a shift from similarity-based retrieval to constraint-aware, cost-faithful, efficient, and set-oriented data systems, providing a concrete path toward realizing our vision.